ALGEBRA 2 · UNIT 9: CONIC SECTIONS

Signals & Dishes

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as engineers for a communications company. They model cell-tower coverage with circles, place a receiver at the focus of a satellite dish, locate the foci of a whispering gallery ellipse, analyze a hyperbolic cooling tower, convert general form to standard form, and design a coverage map.

The reflective properties of conics give the focus a physical meaning: the dish receiver and the whispering spots. Coverage problems turn circle equations into inequalities.

Standards Addressed

- G-GPE.1 Equation of a circle; complete the square
- G-GPE.2 Equation of a parabola from focus and directrix
- G-GPE.3 Ellipses and hyperbolas from foci (Honors)
- A-CED.2 Create equations

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: why are dishes curved? Part 1 circles and parabolas.
Day 2	Part 2 ellipses and hyperbolas.
Day 3	Part 3 coverage map.

Materials

- Student handout
- Graph paper and compass
- String and pins (to draw an ellipse)
- Graphing calculators or Desmos

How the Three Versions Differ

Level	What changes
CP2	Read center and radius, test points, and find a dish focus with $x^2 = 4py$.
CP1	Complete the square, design a dish from diameter and depth, find ellipse foci, and classify general forms.
Honors	Hyperbola from a distance-difference condition, focus-directrix derivation, and an optimized coverage plan.

Watch For (Common Misconceptions)

- Sign errors in the center of $(x - h)^2 + (y - k)^2 = r^2$.
- Using r^2 as the radius.
- Using $c^2 = a^2 + b^2$ for ellipses.

Extension Ideas

- Draw an ellipse with string and two pins.
- Visit a whispering gallery virtually.
- Research how GPS uses intersecting spheres.

Answer Key – CP2

Part 1: Towers & Dishes

#	Task	Answer
1	Find the tower's location (center) and coverage radius.	Center (3, −1), radius 4 miles
2	Is a house at (6, 2) covered? Is one at (5, 1)? Substitute to check.	(6, 2): 9 + 9 = 18 > 16, not covered. (5, 1): 4 + 4 = 8 < 16, covered.
3	A satellite dish has the shape $x^2 = 6y$. For $x^2 = 4py$ the focus is at (0, p). Where should the receiver go?	$4p = 6 \rightarrow p = 1.5$: the receiver goes 1.5 ft above the vertex at (0, 1.5).
4	How deep is the dish if it is 6 ft across (x from −3 to 3)?	$y = 9 \div 6 = 1.5$ ft deep

Part 2: Ellipses & Hyperbolas

#	Task	Answer
5	A whispering gallery has the shape $x^2 \div 2500 + y^2 \div 400 = 1$ (feet). Find a, b, and $c = \sqrt{(a^2 - b^2)}$. Where should two people stand to hear whispers?	$a = 50$, $b = 20$, $c = \sqrt{2100} \approx 45.8$ ft. Stand at the foci (±45.8, 0).
6	A cooling tower's side has the shape $x^2 \div 900 - y^2 \div 1600 = 1$. Find a, b, and $c = \sqrt{(a^2 + b^2)}$.	$a = 30$, $b = 40$, $c = 50$
7	Match each conic to its real-world use and say why the shape is useful.	Dish (parabola) sends signals to its focus; gallery (ellipse) reflects sound between foci; tower (hyperbola) is strong and uses less material.

Part 3: Coverage Map

#	Task	Answer
8	Place one or two towers (radius 5 miles each). Write each equation.	Answers vary.
9	Test every house.	Answers vary.
10	Draw the map.	Answers vary.

Answer Key – CP1

Part 1: Towers & Dishes

#	Task	Answer
1	A tower's coverage boundary is $x^2 + y^2 - 6x + 2y - 6 = 0$. Complete the square to find its center and radius.	$(x - 3)^2 + (y + 1)^2 = 16$: center (3, −1), radius 4
2	Design a dish that is 8 ft across and 1 ft deep. Write its equation $x^2 = 4py$ and find the focus.	At x = 4, y = 1: $16 = 4p \rightarrow p = 4$: $x^2 = 16y$, focus (0, 4)
3	Write the equation of a circle centered at (−2, 4) that passes through (1, 8).	$r^2 = 3^2 + 4^2 = 25$: $(x + 2)^2 + (y - 4)^2 = 25$

Part 2: Ellipses & Hyperbolas

#	Task	Answer
4	For $x^2 \div 2500 + y^2 \div 400 = 1$, find the vertices, co-vertices, and foci.	Vertices (±50, 0); co-vertices (0, ±20); foci $(\pm\sqrt{2100}, 0) \approx (\pm 45.8, 0)$
5	For $x^2 \div 900 - y^2 \div 1600 = 1$, find the vertices, foci, and asymptotes.	Vertices (±30, 0); foci (±50, 0); asymptotes $y = \pm(4 \div 3)x$
6	Classify each: $4x^2 + 9y^2 = 36$; $x^2 - y^2 = 16$; $y = 2x^2 - 3$; $x^2 + y^2 = 25$.	Ellipse; hyperbola; parabola; circle

Part 3: Coverage Map

#	Task	Answer
7	Place the towers and write their equations.	Answers vary. One tower is not enough: from (1.5, 0.5) the houses are about 1.6, 2.9, 6.4, and 6.4 miles away. Two towers work, for example (0, 2) covers (0, 0), (4, 2), and (−3, 5), and (5, −2) covers (6, −4).
8	Show every house is covered.	Answers vary.
9	Explain why your number of towers is the minimum (or near it).	Answers vary.

Answer Key – Honors

Part 1: Definitions

#	Task	Answer
1	Derive the equation of the parabola with focus (0, p) and directrix y = −p.	$\sqrt{(x^2 + (y - p)^2)} = \lvert y + p \rvert \rightarrow x^2 + y^2 - 2py + p^2 = y^2 + 2py + p^2 \rightarrow x^2 = 4py$
2	A dish is 10 ft across and 1.25 ft deep. Find the focus.	x = 5, y = 1.25: 25 = 5p → p = 5; focus at (0, 5), 5 ft above the vertex.
3	Show that every point on $x^2 \div 2500 + y^2 \div 400 = 1$ has distances to the foci that add to 100.	For an ellipse the sum of focal distances is 2a = 100 (shown with a point: at (50, 0), distances 95.8 + 4.2 = 100).

Part 2: Locating a Ship

#	Task	Answer
4	Explain why the ship is on a hyperbola and write its equation.	$\lvert d_1 - d_2 \rvert = 60$ is constant, which defines a hyperbola with foci at the stations: 2a = 60 → a = 30; c = 50; b = 40. $x^2 \div 900 - y^2 \div 1600 = 1$
5	The ship is also on the line y = 32. Find its location (on the right branch).	$x^2 \div 900 = 1 + 1024 \div 1600 = 1.64 \rightarrow x = 30\sqrt{1.64} \approx 38.4$. Location ≈ (38.4, 32).
6	Write the asymptotes and explain what they mean far from the stations.	$y = \pm(4 \div 3)x$; far away the ship's possible positions lie close to these lines.

Part 3: Optimized Coverage

#	Task	Answer
7	Find the smallest single circle that covers all houses, or show two towers are needed with radius 5.	Answers vary.
8	Write the equations and test every house.	Answers vary.
9	Write a brief explaining the design.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Coverage plan uses correct circle equations and point tests, and every conic feature is correct. Goes beyond the minimum with a creative, realistic choice.	Coverage plan uses correct circle equations and point tests, and every conic feature is correct.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Polished, organized, and visually engaging.	Brief includes each conic's equation, key features (center, foci, vertices, radius), and a coverage map. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.